

Final Review

1

- a) Let $p(t)$ be a polynomial with real coefficients. Show that, if $p(\lambda) = 0$ for some $\lambda \in \mathbb{C}$, then $p(\bar{\lambda}) = 0$ as well. Give a few counterexamples to show that the claim fails when p has a coefficient that is not real, i.e. has some nonzero imaginary part.
- b) Let $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be an orthogonal transformation, that is, $A^\top = A^{-1}$. Show that every eigenvalue λ of A satisfies $|\lambda|^2 = \lambda\bar{\lambda} = 1$.
- c) Conclude that if $\det(A) > 0$, then A has a fixed point, i.e. there exists a vector $v \in \mathbb{R}^3$ such that $Av = v$.

2 (Exercise in matrix multiplication)

Let $A \in M_{n \times n}(\mathbb{F})$ be a matrix. Show that $AB = BA$ for every $B \in M_{n \times n}(\mathbb{F})$ if and only if $A = \lambda I$ for some $\lambda \in \mathbb{F}$.

3

Let V, W be vector spaces over a field $\mathbb{F}$. Recall the notation $\mathcal{L}(V, W)$, which is the vector space of all $\mathbb{F}$ -linear maps $V \rightarrow W$.

- a) Carefully construct a linear isomorphism $\phi : \mathcal{L}(\mathbb{R}^n, \mathbb{R}) \rightarrow \mathbb{R}^n$, as $\mathbb{R}$ -vector spaces. (Hint: recall that you can think of linear maps as matrices. What are the dimensions of such matrices in this case?)
- b) Carefully construct a linear isomorphism $\psi : \mathcal{L}(\mathbb{C}^n, \mathbb{C}^m) \rightarrow \mathbb{C}^{m \times n}$, as $\mathbb{C}$ -vector spaces. (Hint: extrapolate from the proof of a.)

4

Let $V \subset \mathbb{C}^n$ be subspace and $P : \mathbb{C}^n \rightarrow V$ be the usual orthogonal projection (i.e. using the Euclidean inner product).

- a) A matrix U (necessarily invertible) is called unitary if $U^* = U^{-1}$. Show that there exists an invertible (**bonus:** unitary) matrix U such that $U^{-1}PU = U^*PU$ is diagonal. Show that $\text{tr}(P) = \text{rank}(P)$.

b) Let P, Q be orthogonal projections onto subspaces of $\mathbb{C}^n$. Show, using part a), that $\text{rank}(P) = \text{rank}(Q)$ holds if and only if there exists some invertible (**bonus:** unitary) matrix $U \in M_n(\mathbb{C})$ such that $UPU^{-1} = Q$.

5 Bonus (Requires knowledge of RREF and row operations)

a) Let $A, B, C \in M_n(\mathbb{F})$ be matrices, and consider the $2n \times 2n$ matrix

$$X = \begin{bmatrix} A & B \\ 0 & C \end{bmatrix},$$

where the notation means that, for instance, the top right $n \times n$ submatrix of X is the matrix B , and so on.

Show that

$$\det(X) = \det(A) \det(C).$$

Hint: start by deriving conditions on A and C to determine whether the RHS is zero or not. Then, break into cases based on these conditions.

b) Give a counterexample using 2×2 matrices to show that the equation

$$\det \left(\begin{bmatrix} A & B \\ C & D \end{bmatrix} \right) = \det(A) \det(D) - \det(B) \det(C)$$

is false in general.

6

Let $0 \neq S \subsetneq \mathbb{R}^n$ be a subspace and P the orthogonal projection onto S . Given a vector $v \in \mathbb{R}^n$, we can write v uniquely as $v = v_{\parallel} + v_{\perp}$, where $v_{\parallel} \in S$ and $v_{\perp} \in S^{\perp}$. The *reflection operator* R_S reflects vectors about the subspace S and is defined by

$$R_S(v) = v_{\parallel} - v_{\perp}.$$

Note that R_S does indeed define a function since $v_{\parallel}, v_{\perp}$ are uniquely defined for any vector v .

a) Prove that $R_S(v) = 2P(v) - v = P(v) - (v - P(v))$. Conclude that $R_S^* = R_S$.

b) What are the eigenvalues of R_S ? (Hint: what are the eigenvalues of P ?)

c) Give an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of R_S (this can be in terms of an ONB for S and $S^{\perp}$).

(Hint: start first by visualizing the problem when $n = 3$ and S is a plane through the origin (say, the xy-plane).

7

Let A be the $n \times n$ matrix whose entries are all 1 (this can be in any field).

Find the eigenvalues and their algebraic/geometric multiplicities (Hint: start by doing rank-nullity to look at the size of the kernel/range. Can you explicitly characterize the range of A ?). Conclude that A is diagonalizable.

8

Diagonalize the $n \times n$ matrix

$$\begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 & 1 \\ 0 & 1 & 0 & \dots & 0 & 1 & 0 \\ 0 & 0 & 1 & \dots & 1 & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 1 & 0 \\ 1 & 0 & 0 & \dots & 0 & 0 & 1 \end{bmatrix}$$

(all 1-s along both diagonals, zeroes elsewhere). (Hint: the argument is different depending on whether n is even or odd. You can find formulae for the characteristic polynomial, but it may be easier to find eigenvectors directly.)

9 6.3.12-13

Let $T : V \rightarrow V$ be a linear transformation on a finite dimensional inner product space.

a) Show that $\text{range}(T^*)^\perp = \ker(T)$ and $\text{range}(T^*) = \ker(T)^\perp$.

b) Show that $\ker(T^*T) = \ker(T)$ and hence $\text{rank}(T^*T) = \text{rank}(T) = \text{rank}(TT^*)$. Is $\text{range}(T^*T) = \text{range}(TT^*)$?